

SUKALYAN ROY

Backend Engineer | Python, Data & ML Systems

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SUMMARY

Backend engineer building data-intensive APIs, time-series storage, streaming paths, and model-serving interfaces in **Python**, **FastAPI**, **PostgreSQL/TimescaleDB**, **Redis**, and **Azure**. Comfortable owning a service from schema and API through deployment and measurement.

PROFESSIONAL EXPERIENCE

Backend Developer | Avinya Neurotech | Full-time, Remote | Jun 2025 to Present

- Reworked a **five-service FastAPI platform**, exposing 50+ REST endpoints for EEG ingestion, analysis, export, and visualization.
- Designed the **TimescaleDB/PostgreSQL schema**, with batched ingestion and time-windowed queries serving live and historical reads over multichannel EEG.
- Built the **bounded WebSocket streaming path**, with fixed-size frames and cancellation of stale requests on seek or disconnect; tuning took first-chunk latency from about 2.2 s to **783 ms p50** and server send time to about **12 ms**.
- Moved exports and frequency-band analysis off the request path with **Redis job queues** and Azure Blob Storage.
- Ran containerized services on **Azure** with CI/CD releases, health checks, versioned database migrations, and **Prometheus/Grafana** observability.

Tech Development Intern | Aecho AI | Fully Remote | Nov 2024 to Jan 2025

- Built a local-first RAG assistant over 300+ pages of internal documentation, cutting routine lookup from minutes to seconds while keeping documents and inference off hosted APIs.

SELECTED PROJECTS

MediPredict ML Microservice & Research Manuscript | FastAPI, scikit-learn, SHAP, Docker

- Sole author of the **FastAPI model-serving service**: typed schemas, artifact-driven model loading, preprocessing shared with training, health and readiness endpoints, SHAP payloads, and Docker packaging; a separate Spring Boot/React application built by the project team consumes it over HTTP.
- Traced a reported R^2 of 0.925 to **target leakage** (a realised-cost field unavailable at quote time) and showed by fixed-model feature ablation that the honest quote-time ceiling sits near R^2 0.15.
- Cut the served artifact from **95.8 MB to 0.39 MB** and single-row inference from about **45 ms to 0.5 ms** at unchanged cross-validated accuracy.

Windowed EEG Signal-Processing Benchmark & Manuscript | Python, NumPy, SciPy, MNE

- Built a reproducible benchmark for windowed filtering, overlap-add, edge policy, and bandpower equivalence on synthetic and clinical EEG data, and documented the failure modes it exposed.
- Measured worst-case p95 filter-plus-feature latency of **6.46 ms** on the 8-channel CHB-MIT case and **1.63 ms** on the 2-channel Sleep-EDF case.

Local-First RAG Chatbot with MCP Integration | Python, FAISS, SQLite, Ollama

- Built a hybrid retrieval pipeline fusing FAISS dense search with BM25 by reciprocal rank fusion, with SQLite keeping retrieval metadata inspectable and a JSON-RPC MCP server serving Streamlit and CLI clients.
- On a fixed 18-query synthetic regression set, hybrid retrieval held Hit Rate@3 at 1.000 and moved MRR from **0.972 to 1.000**; regression evidence, not a general benchmark.

TECHNICAL SKILLS

Primary: Python, SQL | **Working familiarity:** JavaScript, TypeScript, C

Backend and APIs: FastAPI, REST APIs, WebSockets, Node.js, asynchronous job processing

Data Systems: PostgreSQL, TimescaleDB, Redis, Azure Blob Storage, SQLite, FAISS

ML and Retrieval: PyTorch, scikit-learn, SHAP, RAG, Ollama, MCP, NumPy, pandas

Cloud and DevOps: Azure, Docker, GitHub Actions, CI/CD, Prometheus/Grafana, Linux/Bash, Pytest

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Netaji Subhash Engineering College | Kolkata | Graduated 2026 | CGPA: 8.0/10.0

CERTIFICATIONS & ACHIEVEMENTS

Smart India Hackathon National Finalist (2023) | **Google Project Management** | Harvard CS50: CS50x, AI with Python, Cybersecurity